When the roots are available in explicit numeric form, then also we use these convenient notations. We have to be careful when handling roots of higher multiplicity. For instance, if the roots of a cubic equation are 1, 2, 2, then $\sum\alpha = 5$ and $\sum\alpha\beta = (1\times 2)+(1\times 2)+(2\times 2) = 8$.

From the above discussion, we note that for a monic polynomial equation, the sum of the roots is the coefficient of x^{n-1} multiplied by (-1) and the product of the roots is the constant term multiplied by $(-1)^n$.

Example 3.3

If α, β, and γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum\frac{1}{\beta\gamma}$ in terms of the coefficients.

Solution

Since α, β, and γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, we have

$$\sum\nolimits_1 \ \alpha+\beta+\gamma \ = -p \ \text{ and } \ \sum\nolimits_3 \ \alpha\beta\gamma = -r\,.$$

Now

$$\sum\frac{1}{\beta\gamma} \ = \frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}+\frac{1}{\alpha\beta} = \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{-p}{-r} = \frac{p}{r}\,.$$

3.3.2 (c) Formation of Polynomial Equations with given Roots

We have constructed quadratic equations when the roots are known. Now we learn how to form polynomial equations of higher degree when roots are known. How do we find a polynomial equation of degree n with roots $\alpha_1, \alpha_2, \cdots, \alpha_n$? One way of writing a polynomial equation is multiplication of the factors. That is

$$(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n) = 0$$

is a polynomial equation with roots $\alpha_1, \alpha_2, \cdots, \alpha_n$. But it is not the usual way of writing a polynomial equation. We have to write the polynomial equation in the standard form which involves more computations. But by using the relations between roots and coefficients, we can write the polynomial equation directly; moreover, it is possible to write the coefficient of any particular power of x without finding the entire polynomial equation.

A cubic polynomial equation whose roots are α, β, and γ is

$$x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta+\beta\gamma+\gamma\alpha)x - \alpha\beta\gamma = 0\,.$$

A polynomial equation of degree n with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ is given by

$$x^n - \left(\sum\alpha_1\right)x^{n-1} + \left(\sum\alpha_1\alpha_2\right)x^{n-2} - \left(\sum\alpha_1\alpha_2\alpha_3\right)x^{n-3} + \cdots + (-1)^n\,\alpha_1\alpha_2\cdots\alpha_n = 0$$

where, $\sum\alpha_1, \sum\alpha_1\alpha_2, \sum\alpha_1\alpha_2\alpha_3, \ldots$ are as defined earlier.

For instance, a polynomial equation with roots 1, -2, and 3 is given by

$$x^3 - (1-2+3)x^2 + (1\times(-2)+(-2)\times 3+3\times 1)x - 1\times(-2)\times 3 = 0$$

which, on simplification, becomes $x^3 - 2x^2 - 5x + 6 = 0$. It is interesting to verify that the expansion of $(x-1)(x+2)(x-3) = 0$ is $x^3 - 2x^2 - 5x + 6 = 0$.

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